MD. Mehedi Hasan Maruf

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Summary

Software engineer and applied ML researcher. My B.Sc. thesis at Rajshahi University of Engineering & Technology (RUET) built TRACE, which finds the university courses that hold students back by combining prerequisite-graph analysis, Bayesian fail-rate models, and sentiment in student feedback. I then checked its answer with causal inference (DoWhy) on real student records. A paper from it is under review at LAK 2027. I also have 3+ years of industry experience building LLM and RAG features, TypeScript and Python backends, and Docker and CI/CD deployments, currently at Aamira Software Solutions.

Research

TRACE: Curriculum Bottleneck Detection & Causal Validation *Jul 2025 – Aug 2026*

B.Sc. Thesis, Dept. of CSE, RUET

- The question came from my own class. About a third of us failed one first-year mathematics course, while about 4% fail a typical course. Did that course hold students back, or only show who was already behind?
- Built TRACE to find courses like it. It ranks each course by its place in the prerequisite graph (NetworkX, GAT), a Bayesian model of its fail rate, and the tone of student feedback (DeBERTa, VADER).
- Tested TRACE on a synthetic curriculum with known, planted bottlenecks (F-measure 0.75, NDCG@10 0.81). On a real university curriculum, a domain expert agreed with 8 of its top 10.
- Used causal inference to test what drives failure in a flagged module. On 23,369 Open University students (OULAD), a DoWhy backdoor-adjusted model estimated that high assignment engagement raised held-out pass rates by 19.9 percentage points ($p < 0.01$, E-value 1.89).
- After peer review, rebuilt the study into a paper, *When Agreement Misleads: Auditing Course Shortlists in Curriculum Analytics*, now under review at LAK 2027. It gives an exact bound on how much a combined shortlist can change top-k precision. In simulation, sources with a shared bias picked mostly the same courses (88% overlap), yet only 15% of those picks were real bottlenecks.

Experience

Software Engineer II *Jan 2025 – Present*

Aamira Software Solutions, Kokomo, Indiana, USA (remote)

- Raised client response rates by 30% by leading development of a CRM system.
- Cut API response time by 40% by designing and building a TypeScript microservice backend.
- Made the application 60% faster by optimizing its Supabase (PostgreSQL) queries.
- Building a retrieval-augmented generation (RAG) feature for the platform.

Junior Software Engineer *Nov 2022 – Apr 2024*

Code Studio, Rajshahi, Bangladesh

- Kept 12 Linux servers at 99.8% uptime, and cut deployment time by 40% by reworking the CI/CD pipelines.
- Moved 5 client environments onto Docker, which made them simpler to deploy.
- Wrote JavaScript, TypeScript, and Python (Django) backends used by thousands of people a day.

Industrial Trainee *Mar 2025*

Brain Station 23, Dhaka, Bangladesh

- Represented a 15-person trainee cohort to management.
- Built and deployed StudentAI (see Projects), an AI learning assistant on FastAPI and Next.js.

Skills

AI/ML: LLM applications, RAG, LangChain, Hugging Face Transformers (DeBERTa), PyTorch, GNNs (GAT), NLP, causal inference (DoWhy), Bayesian modeling, vector search (pgvector), model evaluation (RLHF)
Languages: Python, TypeScript, JavaScript
Backend: NestJS, FastAPI, Fastify, tRPC, Django
Frontend: Next.js (React), Angular
Databases: PostgreSQL (Supabase), MongoDB, MySQL
DevOps: Docker, GitHub Actions (CI/CD), Linux server administration

Selected Projects

Omnious | Visual Debugging Platform

Nov 2025 – Present

- Building a debugger that draws a program's code graph, animates execution traces, shows where errors cluster, and lets users ask an AI assistant about them.
- A TypeScript Turborepo monorepo: Fastify 5 and tRPC v11 backend, Next.js 16 / React 19 frontend.
- Stores data in PostgreSQL 15 on Supabase, using pgvector, ltree, and pg_trgm for semantic search and tree-shaped data, with row-level security.
- Runs in Docker Compose with multi-stage builds and deploys through GitHub Actions.

StudentAI | Intelligent Learning Companion

Mar 2025

- An AI learning assistant (FastAPI, Next.js) built during industrial training at Brain Station 23.
- Answers questions about uploaded documents through a LangChain RAG pipeline with semantic retrieval and conversation memory; it also analyzes YouTube lectures and writes quizzes.
- Accepts PDFs, images, and video, and has real-time WebSocket chat, JWT login, and progress tracking.
- Deployed on Vercel and Render with a Supabase database.

Education

B.Sc. Engineering, Computer Science and Engineering

Mar 2022 – Aug 2026

Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh

- CGPA: 3.75 / 4.00, Honours

Certifications & Achievements

- Digital Accessibility Training (DAT) — a2i (Aspire to Innovate), Govt. of Bangladesh (Oct 2025)
- Industrial Training Certificate — Brain Station 23 (Mar 2025)
- PHP (Laravel) — EDGE Project (Feb 2025)
- Experience Certificate — Code Studio (Apr 2024)
- 4th Place — RUET CodeSmash, team competitive programming contest (Mar 2023)
- 4th Place — RUET CodeSpark, individual competitive programming contest (Oct 2022)